

Developer Mock Test Results

Student: Nani Lalam

Name:	Nani Lalam	Student ID:	117
Date:	2026-03-05 12:58:16	Performance:	Needs Improvement
Overall Score:	5.0/25 (20.0%)	Test ID:	11e53c88-40ee-4171...
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	1/10	10.0%	Needs Work
MCQ/Theory	4/10	40.0%	Needs Work
Coding	0.0/5	0.0%	Needs Work

Detailed Question Review

APTITUDE SECTION (1/10 - 10.0%)

Q1. A product is on sale for 20% off its original price. If the original price is \$100, how much will you pay for the product after the discount?

Your Answer: \$75

Correct Answer: \$80

The correct answer is \$80 because the discount is 20% of the original price of \$100. The key calculation step is: $\$100 * 0.20 = \20 (discount amount), then $\$100 - \$20 = \$80$ (price after discount). The user's likely mistake was miscalculating the discount amount, as 20% of \$100 is \$20, not \$25, which would be required for the price to be \$75.

Q2. The average of 2 numbers is 18. What is the sum of the 2 numbers?

Your Answer: 32

Correct Answer: 36

The correct answer is 36 because the average of 2 numbers is calculated by dividing their sum by 2. The key calculation step is: $\text{sum of 2 numbers} = \text{average} * 2 = 18 * 2 = 36$. The user's answer of 32 is likely incorrect due to a miscalculation of this step.

Q3. The average of 3 numbers is 15. What is the sum of the 3 numbers?

Your Answer: No answer (Skipped)

Correct Answer: 45

To find the sum of the 3 numbers, we need to multiply the average by the total number of values, which is 3. The key calculation step is: $\text{average} * \text{number of values} = \text{sum}$, so $15 * 3 = 45$. Therefore, the correct answer is indeed 45, and the user happened to choose the correct option, likely by performing this calculation correctly.

■ Q4. A shirt is on sale for 15% off its original price of \$80. How much will you pay for the shirt?

Your Answer: \$65

Correct Answer: \$68

■ To find the sale price, calculate 15% of the original price of \$80: $0.15 * \$80 = \12 . Then, subtract this discount from the original price: $\$80 - \$12 = \$68$. The user's answer of \$65 is likely incorrect because they may have miscalculated the discount or subtracted an incorrect amount from the original price.

■ Q5. What is the average of 2, 4, 6, 8, 10?

Your Answer: 6

■ Excellent! Right answer.

■ Q6. A mixture is made of 2 parts acid and 3 parts water. What is the ratio of acid to the total mixture?

Your Answer: 1/3

Correct Answer: 2/5

■ The correct answer is 2/5 because the mixture is made of 2 parts acid and 3 parts water, making a total of $2 + 3 = 5$ parts. The ratio of acid to the total mixture is then 2/5. The user's likely mistake was calculating the ratio as $2/(2+3) = 2/5$, but incorrectly simplifying it to 1/3, possibly by mistakenly dividing the 2 parts acid by the 3 parts water instead of the total 5 parts.

■ Q7. A store buys a product for \$50 and sells it for \$60. What is the profit percentage?

Your Answer: 15%

Correct Answer: 20%

■ To find the profit percentage, we need to calculate the profit made and then divide it by the cost price. The profit made is $\$60$ (selling price) - $\$50$ (cost price) = $\$10$, and the profit percentage is $(\$10 / \$50) * 100 = 20\%$. The user's answer of 15% is likely incorrect because they may have miscalculated the profit percentage or misunderstood the formula for calculating it.

■ Q8. A mixture is made of 3 parts sugar and 2 parts water. What is the ratio of sugar to water?

Your Answer: 2:3

Correct Answer: 3:2

■ The correct answer is 3:2 because the mixture is made of 3 parts sugar and 2 parts water, directly giving us the ratio of sugar to water. The user's answer of 2:3 is likely incorrect because they may have reversed the given proportions of sugar and water. The key calculation step is simply comparing the given parts: 3 parts sugar / 2 parts water = 3:2 ratio.

■ Q9. A store buys a product for \$40 and sells it for \$50. What is the profit percentage?

Your Answer: 28%

Correct Answer: 25%

■ To find the profit percentage, we need to calculate the profit first: Profit = Selling price - Cost price = $\$50 - \$40 = \$10$. Then, we find the profit percentage: $(\text{Profit} / \text{Cost price}) * 100 = (\$10 / \$40) * 100 = 25\%$. The user's answer of 28% is incorrect, likely due to a miscalculation in the profit percentage calculation.

■ Q10. If it is true that all cars are vehicles and some vehicles are bicycles, what can be concluded?

Your Answer: All bicycles are cars

Correct Answer: Some vehicles are cars

■ The correct answer is "Some vehicles are cars" because we know that all cars are vehicles, which implies that cars are a subset of vehicles. The user's answer, "All bicycles are cars", is incorrect because the statement "some vehicles are bicycles" does not imply that bicycles are cars, but rather that bicycles are a separate type of vehicle. The user likely mistook the relationship between vehicles, cars, and bicycles, incorrectly assuming that bicycles must be cars due to the shared category of vehicles.

■ MCQ SECTION (4/10 - 40.0%)

■ Q1. What type of block is used to handle errors in Java?

Your Answer: Try-catch block

■ Well done!

■ Q2. How should code be organized in Java?

Your Answer: Using methods and variables

Correct Answer: Using classes and packages

■ The correct answer, "Using classes and packages", is right because Java is an object-oriented language that emphasizes organization and structure through classes and packages. This allows for better modularity, reusability, and maintainability of code. The user's answer, "Using methods and variables", is incorrect because while methods and variables are essential components of Java code, they are typically organized within classes and packages to provide a logical and hierarchical structure.

■ Q3. What are methods in Java?

Your Answer: Reusable blocks of code

■ Well done!

■ Q4. What is method overloading in Java?

Your Answer: Multiple methods with the same name but different parameters

■ Correct! Good understanding.

■ Q5. What is the order of operations in Java?

Your Answer: Parentheses, multiplication and division, addition and subtraction, exponentiation

Correct Answer: Parentheses, exponentiation, multiplication and division, addition and subtraction

■ The correct answer is A) Parentheses, exponentiation, multiplication and division, addition and subtraction. This is because Java follows the standard order of operations, often remembered by the acronym PEMDAS (Parentheses, Exponents, Multiplication and Division, and Addition and Subtraction), which ensures that mathematical expressions are evaluated consistently and accurately. The user's answer was incorrect, likely due to mistakenly placing exponentiation after multiplication and division, which could lead to incorrect results in expressions involving these operations.

■ Q6. What can Java be used for?

Your Answer: All of the above

■ Excellent! Right answer.

■ Q7. What is encapsulation in Java?

Your Answer: The concept of hiding all internal details and showing no information

Correct Answer: The concept of hiding internal details and showing only necessary information

■ The correct answer is A) The concept of hiding internal details and showing only necessary information, because encapsulation in Java is a fundamental principle of object-oriented programming that involves bundling data and methods that operate on that data within a single unit, while exposing only the necessary information to the outside world. The user's answer, C) The concept of hiding all internal details and showing no information, is incorrect because encapsulation is not about hiding all internal details, but rather about controlling access to them and exposing only the necessary information. The user's mistake likely stems from a misunderstanding of the concept of encapsulation, which aims to provide a balance between hiding internal implementation details and providing a public interface for interaction.

■ Q8. What happens when a Java file is compiled?

Your Answer:

It is converted to a .java file

Correct Answer:

It is converted to a .class file

■ The correct answer is A) It is converted to a .class file, because when a Java file is compiled, the Java compiler translates the .java file into an intermediate format called bytecode, which is stored in a .class file. This .class file can then be executed by the Java Virtual Machine (JVM). The user's answer, B) It is converted to a .java file, is incorrect because .java files are the source code files that are compiled, not the result of compilation.

■ Q9. What is multi-threading in Java?

Your Answer: The ability of a program to execute multiple processes concurrently

Correct Answer: The ability of a program to execute multiple threads concurrently

■ The correct answer is A) The ability of a program to execute multiple threads concurrently, because multi-threading in Java refers to the ability of a program to execute multiple threads or flows of execution concurrently, sharing the same memory space. This allows for improved responsiveness, efficiency, and system utilization. The user's answer, C) The ability of a program to execute multiple processes concurrently, is incorrect because it describes multi-processing, not multi-threading, which involves separate processes with their own memory spaces.

■ Q10. What is a StackTrace in Java?

Your Answer:

A list of methods that were called after an exception was thrown

Correct Answer:

A list of methods that were called before an exception was thrown

■ The correct answer is A) A list of methods that were called before an exception was thrown, because a StackTrace in Java provides a chronological record of the methods that were invoked leading up to the point where an exception occurred. This allows developers to diagnose and debug issues by tracing the sequence of method calls that resulted in the error. The user's answer, B) A list of methods that were called after an exception was thrown, is incorrect because a StackTrace does not show methods called after the exception, but rather the methods that were on the call stack when the exception was thrown.

■ CODING SECTION (0.0/5 - 0.0%)

■ Q1. Write a Java program that reads an array of integers from stdin and finds the sum of its elements.

****Input Format:** - Line 1: The number of elements...**

Your Answer:

```
public class Main {
    public static void main(String[] args) {
        // Write your solution here
        solution();
    }

    public static void solution() {
        // Your code here
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int sum=0;
        for(int i=0;i<n;i++)
        {
            int sum=sc.nextInt();
            sum=sum+num;
        }
        System.out.println(sum);
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int sum = 0;
        for (int i = 0; i < n; i++) {
            sum += scanner.nextInt();
        }
        System.out.println(sum);
    }
}
```

```

        scanner.close();
    }
}

```

■ The code failed to compile due to a "cannot find symbol" error, indicating that the Java compiler is unable to find the `Scanner` class. This suggests that the `java.util` import statement is missing. The error prevents the code from running, resulting in no output being produced.

■ **Q2. Write a Java program that reads an integer from stdin and calculates its factorial. **Input Format:** - Line 1: An integer **Output Format:** - The ...**

Your Answer:

No answer (Skipped)

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int number = scanner.nextInt();
        long factorial = calculateFactorial(number);
        System.out.println(factorial);
        scanner.close();
    }

    public static long calculateFactorial(int number) {
        long factorial = 1;
        for (int i = 1; i <= number; i++) {
            factorial *= i;
        }
        return factorial;
    }
}

```

■ There is no test case failure to analyze, as no test case results or student code were provided. Without this information, it's impossible to determine what went wrong. Please provide the test case results and student code to receive accurate feedback.

■ **Q3. Write a program that reads a list of integers and prints the list with all duplicates removed. **Input Format:** - Line 1: Number of integers (integers...**

Your Answer:

```

import java.util.*;
public class Main{
    public static void main(string[] args){
        solution();
    }
    public static void solution(){

        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        LinkedHshSet<Integer> set=new LinkedHashSet<>();
        for(int i=0;i<n;i++){
            set.add(sc.nextInt());
        }
        for(int i=0;i<n;i++){
            set.add(sc.nextInt());
        }
        for(int num:set){
            system.out.println(num+" ");
        }
    }
}

```

```
}  
}
```

Correct Answer:

```
import java.util.Scanner;  
  
public class Main {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
        int n = scanner.nextInt();  
        scanner.nextLine(); // Consume newline left-over  
        String[] input = scanner.nextLine().split(" ");  
        int[] numbers = new int[n];  
        for (int i = 0; i < n; i++) {  
            numbers[i] = Integer.parseInt(input[i]);  
        }  
        scanner.close();  
  
        // Remove duplicates  
        java.util.Set<Integer> set = new java.util.HashSet<>();  
        for (int num : numbers) {  
            set.add(num);  
        }  
  
        // Print the result  
        for (Integer num : set) {  
            System.out.print(num + " ");  
        }  
    }  
}
```

■ The code failed to compile due to a symbol error, specifically it cannot find the class "string" which should be "String" with a capital 'S'. This typo in the main method parameter type caused the compilation error. The actual output was empty because the code did not run successfully.

■ **Q4. Write a Java program that reads an integer from stdin and checks if it is a Fibonacci number. **Input Format:** - Line 1: An integer **Output Format...**

Your Answer:

No answer (Skipped)

Correct Answer:

```
import java.util.Scanner;  
  
public class Main {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
        int n = scanner.nextInt();  
        boolean isFibonacci = isPerfectSquare(5 * n * n + 4) || isPerfectSquare(5 * n * n - 4);  
        System.out.println(isFibonacci ? "Fibonacci" : "Not Fibonacci");  
        scanner.close();  
    }  
  
    public static boolean isPerfectSquare(int x) {  
        int s = (int) Math.sqrt(x);  
        return s * s == x;  
    }  
}
```

■ There is no test case failure provided to analyze. The student code is also not submitted, so it's impossible to determine what went wrong. Please provide the test case results and the student's code to give accurate feedback.

■ Q5. Write a Java program that reads a string and a substring from stdin and checks if the string contains the substring. ****Input Format:**** - Line 1: The ...

Your Answer:

No answer (Skipped)

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String str = scanner.nextLine();
        String substr = scanner.nextLine();
        if (str.contains(substr)) {
            System.out.println("Contains");
        } else {
            System.out.println("Not Contains");
        }
        scanner.close();
    }
}
```

■ There is no test case failure shown, as no test case results are provided. The input is given, but the expected output and actual output are not shown, making it impossible to determine what went wrong. Without the actual test case results, it's not possible to provide accurate feedback.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.